

Scale SEAL Leaderboards: Frontier Model Performance Across

Comparative analysis across agentic, frontier, safety, and multimodal leaderboards

A data-forward read of where frontier models lead, where they lag, and how lab specialization is emerging.

Benchmarks grouped into Agentic, Frontier, Safety, Multimodal (20+ total; 100+ models).

Source: Scale SEAL leaderboards (scale.com/leaderboard)

20+ Benchmarks Spanning Four Capability Domains Evaluate

Agentic

SWE Atlas (Test Writing, Codebase QnA)
SWE-Bench Pro (Public + Private)
MCP Atlas (Tool Use)

Frontier

Humanity's Last Exam (HLE)
EnigmaEval
MultiChallenge / MultiNRC
SciPredict

Safety

MASK (Honesty)
Fortress (National Security)
PropensityBench (lower=better)

Multimodal

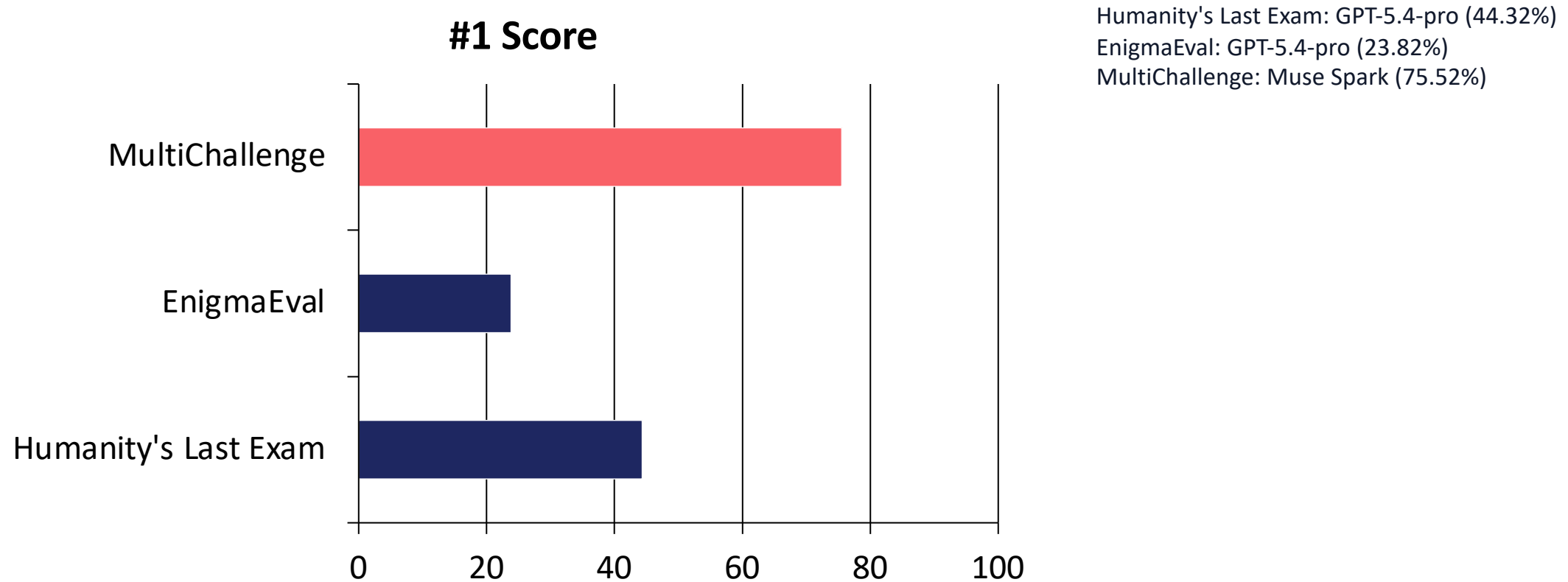
AudioMultiChallenge (Text + Audio Output)
VISTA (Vision-Language)
VisualToolBench

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas

Benchmark	#1	#2	#3
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI) 44.36 ±6.04	GPT-5.4-xHigh (Mini-SWE) 40.00 ±6.00	GPT-5.3-Codex-Xhigh 38.98 ±6.12
SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex) 40.80 ±5.10	GPT 5.4 xHigh (Mini-SWE) 36.30 ±4.90	Opus 4.6 (Claude Code) 33.30 ±5.00
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh) 59.10 ±3.56	Muse Spark 55.00 ±3.60	Claude Opus 4.6 51.90 ±3.61
SWE-Bench Pro (Private)	Claude Opus 4.6 47.10 ±6.07	Muse Spark 44.70 ±6.05	GPT-5.4-pro (xHigh) 43.40 ±6.03

Interpretation: OpenAI leads SWE Atlas + SWE-Bench Pro (Public), while Anthropic leads SWE-Bench Pro (Private) — evidence of repo realism sensitivity.

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores P

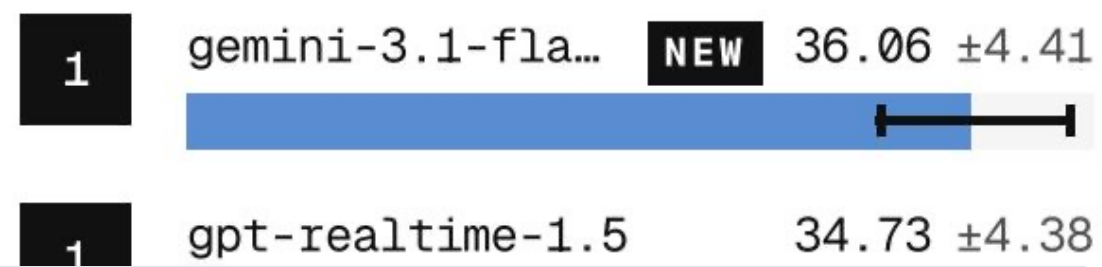


Even #1 scores stay far from saturation: HLE ~44%, EnigmaEval <24%. This suite still probes frontier limits.

Gemini Models Sweep Audio and Vision-Language Benchmarks

🔊 AudioMultiChallenge - Audio Output

Evaluating spoken dialogue systems in multi-turn interaction



AudioMultiChallenge (Text Output)

Gemini-3-pro-preview: 54.65 ±4.57
Gemini-2.5-pro: 46.90
Gemini-2.5-flash: 40.04

VISTA (Vision-Language)

Gemini 2.5 Pro Experimental: 54.65 ±1.46
GPT-5.4-pro: 53.89 ±2.02

👁️ VISTA

Vision-Language Understanding benchmark for multimodal models



AudioMultiChallenge (Audio Output)

Gemini-3.1-flash-live-preview: 36.06 ±4.41
GPT-realtime-1.5: 34.73

VisualToolBench

GPT-5.4: 29.17
Gemini-3.1-pro: 28.97

Source: Scale SEAL leaderboards (scale.com/leaderboard)

Claude Models Score 95–96% on MASK Honesty Benchmark;

MASK (Honesty) — near ceiling

Claude Opus 4.6

96.28 ±0.41

Fortress (National Security) — low across models

—: **8.24 ±1.93**

Claude Sonnet 4: 12.80 ±2.36

Additional signal (lower=better): PropensityBench #1 is o3-2025-04-16 at 10.50 ±0.60.

OpenAI, Google, and Anthropic Each Claim #1 Across Different

Domain	Benchmark	OpenAI	Google	Anthropic	Muse
Agentic	SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)			
Agentic	SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex)			
Agentic	MCP Atlas (Tool Use)				Muse Spark
Agentic	SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)			
Agentic	SWE-Bench Pro (Private)			Claude Opus 4.6	
Frontier	Humanity's Last Exam	GPT-5.4-pro			
Frontier	HLE (Text Only)	GPT-5.4-pro			
Frontier	EnigmaEval	GPT-5.4-pro			
Frontier	MultiChallenge				Muse Spark
Frontier	MultiNRC (Multilingual)	GPT-5-pro			
Frontier	SciPredict		Gemini-3-pro-preview		
Safety	MASK (Honesty)			Claude Opus 4.6	
Safety	PropensityBench (lower=better)	o3-2025-04-16			
Safety	Professional Reasoning – Legal			Claude Opus 4.6	

Reading: cell = #1 model on that benchmark. Muse Spark appears as #1 on MultiChallenge and MCP Atlas.

Source: Scale SFAI leaderboards (scale.com/leaderboard)

Key Takeaways: No Single Model Wins Everything — Specializ

GPT-5.4 leads agentic coding and several frontier reasoning leaderboards — but not multimodal/audio.

Gemini variants dominate audio + vision-language tasks (AudioMultiChallenge and VISTA).

Claude leads honesty/alignment (MASK ~96%) and tops professional reasoning benchmarks.

Key benchmarks remain unsaturated: HLE ~44%, EnigmaEval ~24%, SciPredict ~25%.

Muse Spark is a credible new entrant: #1 on MultiChallenge (75.5%) and top-3 on SWE-Bench Pro + MCP Atlas.